

MANAGEMENT PLAN FOR MARINE BIOTOXINS IN CALIFORNIA SHELLFISH

California Department of Public Health
Environmental Management Branch
Preharvest Shellfish and Marine Biotoxin Monitoring Program

2025

EXECUTIVE SUMMARY

This management plan for marine biotoxins in California shellfish is prepared in compliance with the National Shellfish Sanitation Program (NSSP) Model Ordinance (2023). This document will be revised as necessary in response to new data and experience in the management of biotoxin events.

The California Department of Public Health (CDPH) is the lead agency in the state Marine Biotoxin Monitoring Program, an important part of the State shellfish sanitation program. Within the CDPH Center for Environmental Health (CEH) and the Division of Radiation Safety and Environmental Management, the Environmental Management Branch (EMB) regulates shellfish sanitation in the certified commercial growing areas prior to harvest. The Food and Drug Branch, within the Division of Food, Drug, and Radiation Safety, regulates shellfish sanitation after harvest. EMB manages the Marine Biotoxin Monitoring Program for bivalve shellfish in California. Other CDPH agencies participating in the program are the Division of Communicable Disease Control, the Microbial Diseases Laboratory, the Food and Drug Laboratory, and the Office of Public Affairs. Also participating, through a memorandum of understanding with CDPH, is the Department of Fish and Wildlife, which provides assistance upon request in the areas of sampling, public information, and patrol and enforcement.

Bivalve shellfish in California (especially mussels and oysters) have long been known to be susceptible to contamination with the toxins that cause paralytic shellfish poisoning (PSP). In the fall of 1991, another natural neurotoxin called domoic acid was discovered in the California marine environment in several species, including shellfish. Domoic acid poisoning involves a wide range of symptoms, including severe neurological effects that have been referred to as amnesic shellfish poisoning (ASP). Although ASP is not an accurate term for the majority of identified symptoms, it has been in common usage since the first documented outbreak in Prince Edward Island in 1987 and will be used in the current document. A third natural toxin group that causes diarrhetic shellfish poisoning (DSP) has been detected below the alert level, by CDPH in partnership with FDA, in mussel samples on two occasions in 2013 and 2018.

Routine PSP prevention activities consist of an annual quarantine on sport-harvested mussels established every year from May through October, and an ongoing public education program largely keyed to the annual mussel quarantine. In addition, there is mandatory disease reporting for PSP in California. An extensive early warning system is in place that includes required sampling of commercially harvested shellfish, a coastal mussel sampling program at key stations, an innovative volunteer-based phytoplankton monitoring program, and use of satellite imagery and oceanographic data from offshore buoys. These elements are intended to provide an early warning for toxigenic plankton blooms that could result in PSP, ASP, or DSP in shellfish consumers.

Initial response actions to detection of toxin(s) above the alert level are increased sampling at key stations and additional secondary stations, as well as notification of local health authorities. Control actions directed toward commercial harvesters consist

of harvest closures and, if necessary, tracing, embargo and recall of harvested product. Control actions directed toward recreational sport-harvesters consist of quarantines, health advisories, and expedited public information measures such as press releases, a toll-free recorded message, and web-based materials. This plan establishes criteria for sampling, harvest closures and quarantines, harvested product actions, enforcement and patrol, and reopening.

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I. BACKGROUND

California is a participant in the National Shellfish Sanitation Program (NSSP), a voluntary cooperative program consisting of three partners: state shellfish regulatory agencies, federal agencies, particularly the U.S. Food and Drug Administration (FDA), and the shellfish industry.

The Interstate Shellfish Sanitation Conference (ISSC), of which California is a member, establishes uniform guidelines for the NSSP. The ISSC adopts the NSSP Model Ordinance, as a uniform guide for the states to use for exercising supervision over shellfish growing and relaying areas, and for issuing certificates to shellfish shippers. FDA also uses the Model Ordinance for evaluating State shellfish sanitation programs to determine if the programs conform to the recommended guidelines of the NSSP.

The Model Ordinance requires the state shellfish authority to establish a biotoxin management plan for specific biotoxins that meet the following criteria: if there has been a history of biotoxin closures; if toxin-producing phytoplankton are known to occur in the growing area; or a reasonable likelihood that biotoxin closures could occur. This biotoxin management plan covers procedures for response to paralytic shellfish poisoning, domoic acid, and diarrhetic shellfish poisoning toxins. This management plan is issued in compliance with that requirement, and follows the guidelines in the NSSP Model Ordinance, Chapter IV (2023).

1. Paralytic Shellfish Poisoning

Paralytic shellfish poisoning (PSP) is an acute, potentially fatal form of food poisoning that is associated with the consumption of bivalve molluscs (for example, mussels, oysters, clams, and scallops) that have fed on the toxin-producing dinoflagellate *Alexandrium catenella*. Eating shellfish that contain PSP toxins leads to gastrointestinal distress and acute neurologic symptoms within a few minutes to a few hours. The PSP toxins, saxitoxin and its numerous analogs, are sodium channel blockers which inhibit neural transmission. Neurologic symptoms begin with tingling and numbness of the lips, tongue, and fingertips, followed by disturbed balance, lack of muscular coordination, slurred speech and difficulty in swallowing. In severe poisoning, complete muscular paralysis and death from asphyxiation can occur if breathing is not maintained by artificial means. There is no known antidote to the poison. Symptoms tend to resolve entirely in a day or two under proper medical care. Persons who suspect they or others are experiencing PSP symptoms should immediately seek medical treatment.

The type and severity of symptoms depends on the amount of toxic shellfish consumed as well as the specific toxicity of the shellfish. Price et al. (1991) summarize the range of toxin dose responses as follows: 200 to 500 micrograms (μg) will cause at least minor symptoms, 500 to 2000 μg will cause moderate to severe symptoms, and toxin concentrations greater than 2000 μg will produce serious to lethal effects. It should be noted that exceptions exist and serious health effects have also been documented at much lower concentrations (100 to 400 μg). The federal alert level for PSP toxicity is 80

µg per 100 grams (g) of shellfish tissue, and the detection limit for the PSP mouse bioassay is approximately 32 µg/100 g.

In California, PSP occurs most commonly during the warm spring, summer, and early fall months, although episodes of high toxicity in shellfish have occurred during the winter months. PSP was made a reportable disease in 1927. There is some limited information about cases prior to 1927 although there could be underreporting during this time. Since 1903, 582 total cases including 543 illnesses and 39 deaths have been attributed to PSP in California. Over 99 percent of these cases have occurred during the months of May through October. The last major PSP outbreak in California occurred in July 1980 with 98 cases and 2 deaths. In August 1991, 11 non-fatal cases, including 3 that were hospitalized, were reported in persons who had eaten mussels they had collected in northern Sonoma County. In March 2018, 1 non-fatal case including hospitalization was reported in a person who ate mussels sport-collected in northern Marin County. Elevated levels of the PSP toxins have historically been detected in mussels, oysters, scallop viscera, clams, abalone jingles, and gooseneck barnacles.

Alexandrium is a naturally occurring unicellular algae that is normally absent but occasionally can constitute a minor component of the marine phytoplankton community along the California coast. Under favorable environmental conditions this dinoflagellate may undergo periods of rapid population growth, frequently referred to as a “bloom”. The term “bloom” or “red tide” is misleading with respect to *Alexandrium* and the resultant PSP toxicity in shellfish. Visible blooms of *Alexandrium* are rarely seen along the California coast. Conversely, elevated levels of PSP toxins in shellfish can result from the presence of relatively low numbers (e.g., 100 cells per liter) of *Alexandrium* in the water. In addition, there can be tremendous variability in the rate of toxin uptake, concentration, and depuration among different species of bivalve shellfish. The concentration of toxin present in the shellfish is therefore dependent on the number of toxin-producing cells in the water as well as on the species of shellfish.

The source of the dinoflagellates that provide the “seed” for such blooms is in question, but two likely scenarios are responsible. First, resting cysts of *Alexandrium* in local sediments can, under favorable conditions, produce vegetative cells that can then reproduce both sexually and asexually, resulting in localized “hot spots” of PSP toxicity in shellfish. Second, this dinoflagellate may be transported in offshore warm water masses that can move onshore under certain environmental conditions such as downwelling. This advection process could potentially result in either a quick spike in PSP toxicity if the number of transported cells is high, or it may simply provide the cells necessary for a bloom to initiate. Regardless of the origins of the toxin-producing dinoflagellates, the general pattern has been for these blooms to be detected first along the open coast, occasionally followed by transport into bays and estuaries. The degree to which coastal phytoplankton blooms intrude into bays and estuaries is likely influenced in part by the orientation of the bay relative to coastal currents and the extent of tidal mixing and transport that occurs. Thus, the PSP management program that evolved in California includes large elements of surveillance and rapid response.

Historical data from California's monitoring program indicate several patterns with respect to the rate and magnitude of PSP toxicity, as well as the spatial and temporal distribution, along the coast. For the purpose of this discussion a PSP event is considered to be an occurrence of a PSP toxin concentration in shellfish that exceeds the federal alert level at one or more sampling stations. PSP events may affect a relatively discreet section of coastline (e.g., the Marin coast south of Pt. Reyes and north of Muir Beach) or they may affect a widespread area of the coast (i.e., the impact may extend over several coastal counties). The majority of widespread events have exhibited a pattern of rapid toxin increase, often going from nondetectable levels to those associated with varying degrees of illness within several days. Unfortunately, the early detection of low toxin levels does not provide enough information regarding the likelihood of a major event. There are frequent occurrences of a pattern in which low-level toxicity persists for several weeks in an area but does not progress above the alert level. Additional information such as oceanographic conditions (sea surface temperature, wind speed and direction, upwelling indices) and phytoplankton species composition and abundance, in combination with toxin monitoring, can provide better insight into the potential for the development of a major PSP event. For a detailed review of the history of PSP in California and the development of the monitoring program, see Price et al. (1991).

2. Domoic Acid

In October of 1991 the presence of another marine biotoxin was confirmed in California's coastal waters. Domoic acid toxicity, which can result in the condition called amnesic shellfish poisoning (ASP), was identified as the cause of death in a large number of brown pelicans and Brandt's cormorants in the Santa Cruz area of Monterey Bay (Work, 1993). The birds had been feeding on schools of anchovies (*Engraulis mordax*) in the bay, which in turn had been feeding on a bloom of the diatom *Pseudo-nitzschia australis*.

The only documented domoic acid event prior to 1991 was a serious episode in Prince Edward Island, eastern Canada, in 1987 in which three people died and over 100 people were made ill from the consumption of toxic mussels. Domoic acid is a neuroexcitatory amino acid and thus results in over-stimulation of certain nerves cells in the brain, potentially with permanent or fatal effects. Case studies of the Canadian episode indicated that the most common symptoms were gastrointestinal, followed by neurologic symptoms including headaches, loss of balance and/or dizziness, memory loss, varying degrees of confusion, disorientation, changes in the level of consciousness, and in some cases seizures (Teitelbaum, 1990; Perl et al., 1990). There have been no reported cases of human illness related to domoic acid from shellfish harvested in California.

Based on the rather small number of case histories available the following dose responses can be approximated while recognizing the overlap in ranges and symptoms: 27 to 75 µg/g may result in mild to moderate symptoms (gastrointestinal), 40 to 700 µg/g may result in moderate to severe neurologic symptoms, and domoic acid

concentrations greater than 450 µg/g may result in severe neurologic symptoms and/or death. It should again be noted that there were exceptions to these estimates in the Canadian event, as serious health effects were documented in several patients at low toxin concentrations (i.e., < 40 µg/g). The federal alert level for domoic acid is 20 µg per gram (g) of shellfish tissue or 20 parts per million (ppm), and the detection limit for the chemical analysis is approximately 2.5 µg/g.

The first domoic acid event documented in California in 1991 had a maximum detected concentration of 47 µg/g in mussels from Point Pinos (Monterey County). In the years immediately following this event the level of domoic acid detected in shellfish was low (<20 µg/g) or nondetectable despite the observation of periodic *Pseudo-nitzschia* blooms along the coast. The relative lack of toxicity in mussels in the presence of high densities of toxic *Pseudo-nitzschia* raised questions about the suitability of this organism for the early detection of domoic acid. In the winter and early spring of 2002, a major domoic acid event swept down the coast from Monterey Bay to Los Angeles. Record levels of domoic acid were detected in mussels from all counties in that region: Santa Cruz (120 µg/g), San Luis Obispo (113 µg/g), Santa Barbara (380 µg/g), Ventura (56 µg/g), and Los Angeles (170 µg/g). It is evident that mussels along the California coast can accumulate significant levels of domoic acid and are therefore a concern with respect to public health protection. The rate of toxin increase in mussels and oysters can be quite rapid in some cases, increasing from less than 1 µg/g to 56 µg/g in one week inside Morro Bay in 2002 and increasing from < 2.5 µg/g to 170 µg/g in one week at a Santa Barbara site in 2015. Toxin increase can be quite rapid and that shellfish monitoring at a frequency of less than one sample per week may not be effective in providing early detection of domoic acid. Weekly phytoplankton monitoring has proven to provide an advance warning of a potential *Pseudo-nitzschia* bloom.

3. Diarrhetic Shellfish Poisoning

Diarrhetic shellfish poisoning (DSP) is an acute form of food poisoning that is associated with the consumption of bivalve molluscs (for example, mussels, oysters, clams, and scallops) that have fed on the toxin-producing dinoflagellate *Dinophysis*. The DSP toxin is non-lethal to humans. Several species of the dinoflagellate genus *Dinophysis* are known to produce a number of fat-soluble toxins collectively referred to as diarrhetic shellfish poisoning (DSP) toxins. These include okadaic acid (OA), dinophysistoxin-1 (DTX-1) and dinophysistoxin-2 (DTX-2).

DSP is primarily observed as a mild gastrointestinal disorder. Symptoms may begin within 30 minutes to 3 hours and last up to 2 or 3 days. They may include nausea, vomiting, diarrhea, and abdominal pain accompanied by chills, headache, and fever. There have been no reported cases of illness related to DSP from shellfish harvested in California.

The federal alert level for DSP toxicity is 0.16 milligrams per kilogram (mg/kg) or 0.16 ppm of shellfish tissue. The NSSP approved method for DSP is a Liquid Chromatography Tandem Mass Spectrometry (LC-MS/MS) method. The method

specifically measures the suite of lipophilic toxins associated with DSP: okadaic acid (OA), dinophysistoxin 1 (DTX-1), and dinophysistoxin 2 (DTX-2). The combined total DSP toxins are referred to as diarrhetic shellfish toxins (DST). The method is approved for use in clams and approved limited use for mussels and oysters.

Dinophysis is a naturally occurring unicellular algae that is normally absent but occasionally can constitute a minor component of the marine phytoplankton community along the California coast. There have been very few occasions where elevated concentrations of *Dinophysis* were observed. These events occurred in May 2013 and July 2018 at the Cal Poly Pier in San Luis Obispo. Additionally, DSP toxin levels greater than 0.16 ppm OA-equivalents have been found in sentinel mussels from Santa Cruz in Monterey Bay in 7 out of 190 samples collected from 2013 through 2016 (Shultz 2019).

In 2013, high cell densities (> 50,000 cells/L) of *Dinophysis* spp. were reported at the Cal Poly Pier in Avila Bay on May 12. Phytoplankton samples collected by the CDPH network from surrounding sites in San Luis Obispo County did not reveal high numbers of this dinoflagellate, although several species of *Dinophysis* were present in low numbers. CDPH collected mussel samples at the Cal Poly Pier, from outer Morro Bay, and at Goleta Pier in Santa Barbara. The latter location had the greatest numbers of *Dinophysis* observed in CDPH samples, although these were still exceedingly low ($\leq 2\%$ species composition) compared to the other species present. CDPH also collected phytoplankton samples (concentrated net tow and whole water sample composited over three depths) from the Cal Poly Pier and local rock crab. All samples were analyzed by LC-MS/MS at the FDA's Gulf Coast Seafood Laboratory in Alabama. The FDA laboratory reported the detection of trace levels of DSP toxins in all mussel and plankton samples, all of which were well below the 0.16 ppm regulatory threshold for DST. These toxins were not detected in the crab viscera. The highest toxin concentration, 0.04 ppm, was found in the Goleta Pier mussels, corresponding to our phytoplankton observations of higher *Dinophysis* numbers at this location. All three samples of mussels, the concentrated net tow, and the composited whole water sample contained OA and DTX-1; DTX-2 was not found in any sample. Subsequent phytoplankton samples from the affected region did not reveal elevated numbers of *Dinophysis*, therefore additional sampling for DSP analysis was not warranted.

In 2018, elevated levels (11,195 cells/L) of *Dinophysis* were reported at the Cal Poly Pier on June 25. CDPH coordinated sample collection of mussels from Cal Poly Pier, outer Morro Bay, and Montana De Oro State Park, and oysters from the Morro Bay growing area. The samples were analyzed by LC-MS/MS at the FDA laboratory in College Park, Maryland. DST were highest in the Cal Poly Pier sample at 0.033 ppm and at trace levels in the other mussel samples. DST was not detected in the oyster sample. As in the 2013 sampling, the Cal Poly Pier mussel sample contained OA and DXT-1, but not DTX-2. The Morro Bay and Montana De Oro State Park mussel samples contained trace amounts of DXT-1 only. Subsequent phytoplankton samples did not reveal elevated numbers of *Dinophysis*, therefore additional sampling for DSP analysis was not warranted.

4. Plankton

The marine biotoxins that are known to exist along the California coast originate from a small number of phytoplankton species, as previously described. The cost and time involved in routinely conducting separate analyses for each toxin on every shellfish sample makes this approach to biotoxin monitoring impractical. It is often difficult to achieve an adequate frequency of coastal shellfish sampling to ensure the early detection of a potential biotoxin event. In addition, other known toxins may be present along the California coast that would not be detected with the current methods for domoic acid analysis or PSP toxin assay (e.g., diarrhetic shellfish poisoning, or DSP). A review of the toxin-producing phytoplankton and their known distributions can be found in Shumway (1991).

CDPH Marine Biotoxin Monitoring Program monitors phytoplankton as a cost-effective early warning system. The intent of this program is to develop a network of volunteer samplers and field observers that would allow the early detection of potentially toxigenic blooms. Early detection is key to mobilizing and focusing additional sampling and analytical resources for plankton, shellfish, and other species in the affected region. As a result of this volunteer effort CDPH has been able to detect and track numerous harmful algal blooms, improving our ability to protect the public's health. This approach to biotoxin management allows CDPH to comply with the NSSP Section II Chapter IV @.04 B, which states that phytoplankton monitoring can be used in combination with shellfish toxicity screenings.

The phytoplankton monitoring efforts to date have allowed the Marine Biotoxin Monitoring Program to improve its understanding of the spatial and temporal distribution of phytoplankton assemblages, in particular the known toxin producers, along the California coast. For example, observations of very low numbers of *Alexandrium* often correspond to low levels of PSP toxins, just above the detection limit. Routine monitoring has shown that Monterey Bay and the southern California coast, from San Luis Obispo through Ventura counties, experience the most intense *Pseudo-nitzschia* blooms, with Santa Barbara County having the highest domoic acid levels detected in mussels and lobster viscera.

5. Program Components

The Marine Biotoxin Monitoring Program has historically been comprised of five main elements, with a sixth element added in response to the identification of domoic acid in California shellfish: (1) an annual quarantine, established from May through October, on sport-harvesting of mussels along the entire California coastline; (2) public education activities; (3) weekly testing of shellfish from each commercial shellfish growing area; (4) coastal monitoring of shellfish; (5) mandatory reporting of any known or suspected case of PSP or ASP to local health authorities, who inform CDPH; and (6) a volunteer-based phytoplankton monitoring program for early detection of toxic species. Data from coastal buoys (e.g., wind speed and direction, sea surface temperatures) and occasionally satellite imagery (e.g., sea surface temperature imagery, primary

productivity imagery) provides useful near real time information on environmental trends. Response actions (e.g., follow-up sampling, special quarantines on sport-harvested shellfish, closure of commercial shellfish growing areas, and recalls or other harvested product actions) are ordered when needed.

All commercial shellfish harvesting in California involves shellfish that are cultivated by certified harvesters on well-defined and typically small-acreage aquaculture leases, which are located mostly within bays and estuaries (Appendix A). No wild shellfish are harvested from natural beds.

Before harvesting shellfish for human consumption, each shellfish harvester must obtain a shellfish growing area certificate from CDPH. A certificate is valid only for the defined area under the control of the harvester to whom it is issued. Because of this distinction between commercial shellfish harvesting areas and other areas where naturally-occurring shellfish may be taken by sport harvesters, the California Marine Biotoxin Monitoring Program regulates these two types of areas differently, as described in this management plan.

II. PROGRAM ORGANIZATION AND RESPONSIBILITIES

1. Department of Public Health (CDPH)

The marine biotoxins monitoring program involves many diverse units within CDPH, most of which are the responsibility of the Deputy Director of the Center for Environmental Health. The major responsibilities for the program are assigned as follows (see Appendix B for a staff roster):

- a. *Center for Environmental Health (CEH),
Division of Radiation Safety & Environmental Management (DRSEM),
Environmental Management Branch (EMB),
Environmental Health Services Section (EHSS).*

The chief of the Preharvest Shellfish Unit (PSU) within EHSS coordinates the management of the Marine Biotoxin Monitoring Program for bivalve shellfish and reports to the Chief of EHSS. PSU coordinates routine sampling and analysis of phytoplankton for harmful algal bloom species and bivalve shellfish for PSP and ASP toxins. CDPH PSU headquarters and phytoplankton laboratory is located at 850 Marina Bay Parkway, Richmond, CA.

- b. *Division of Food, Drug, and Radiation Safety (DFDRS).*

(1) Food and Drug Branch (FDB)

The Chief of this Branch is responsible for the postharvest shellfish sanitation program, as it relates to the regulation of shellfish handling, processing, storage and distribution after harvest, including the labeling and identification of shellfish moving in commerce.

The Chief of the Food Safety Section coordinates the postharvest program within this Branch. FDB is responsible for collecting samples of commercial seafood species (e.g., crab, sardine, and anchovy) when EMB's environmental monitoring detects increases in toxin-producing phytoplankton or in toxins in routine shellfish samples. Shellfish sanitation activities are included with the many other responsibilities of FDB's staff of food and drug investigators.

(2) Food and Drug Laboratory Branch (FDLB)

The Chief of FDLB manages a staff of analytical chemists that perform the analysis of shellfish and other seafood samples for domoic acid, and reports the laboratory results immediately to relevant staff within EMB and FDB. CDPH FDL is located at 850 Marina Bay Parkway, Richmond, CA

c. *Division of Communicable Disease Control* (DCDC) under the *Center for Communicable Disease Control*

(1) Disease Investigations and Surveillance Branch, Disease Investigations Section (DIS)

The Chief of DIS is responsible for reviewing foodborne disease reports, for conducting epidemiological and other follow-up investigations of cases, and providing information on any shellfish poisoning cases to the Chief of the Food Safety Section, FDB, and to the PSU Chief as appropriate.

(2) Microbial Diseases Laboratory (MDL)

The Chief of MDL is responsible for performing laboratory assays of shellfish for paralytic shellfish poisoning toxins, and for reporting laboratory results immediately to the PSU staff. CDPH MDL is located at 850 Marina Bay Parkway, Richmond, CA.

d. *Office of Communication* (COMMS)

Immediately subordinate to the Director of Public Health, the Chief of COMMS (previously known as Office of Public Affairs or OPA) is responsible for disseminating press releases pertaining to shellfish poisoning and other issues, and for managing contacts between the news media and CDPH staff.

2. Department of Fish and Wildlife (DFW)

DFW participates in the Marine Biotoxin Monitoring Program through the terms of the "Memorandum of Understanding (MOU) between the Department of Fish and Game and the Department of Public Health regarding the California Shellfish Sanitation and Marine Biotoxins Monitoring and Control Program, signed in June 2016 (Appendix C).

a. *Wildlife Protection Division*

This Division, with its game wardens, provides assistance to the Marine Biotoxin Monitoring Program upon request in the areas of public information and patrol.

b. Marine Resources Division

This Division, with its marine biologists, also is available to provide assistance, particularly in the area of shellfish sampling, if requested by CDPH during a toxic phytoplankton bloom or shellfish poisoning emergency.

3. Local Health Agencies

The local health agencies of the State, consisting of 57 county health departments, one city-county department (San Francisco), and four city health departments, are a key element in protecting the public from shellfish poisoning.

a. Local Health Officers

The local health officer, a physician, is the person responsible for all public health activities within each local health agency jurisdiction. A key responsibility of the health officer is to maintain communications with the medical community, especially notifying hospital emergency rooms and poison control centers when a shellfish poisoning alert is received.

b. Directors of Environmental Health (DEH)

Forty-eight counties and the four cities that have health departments have a director of environmental health. The remaining 10 counties contract for environmental health services from CDPH. In most cases the DEH heads an environmental health division within a local health department under the direction of the health officer, although in some cases the DEH is the head of a separate department and/or subordinate to some other official. In any case, the 15 coastal environmental health agencies post quarantine notices or provide posters to other agencies within their area of jurisdiction, such as parks, for posting. In addition, in coordination with EMB and as their resources allow, they collect and ship shellfish samples as participants in the Marine Biotoxin Monitoring Program. Local health departments are also responsible for investigating cases of shellfish poisoning reported by providers or the public and reporting the cases to CDPH DIS.

III. ROUTINE PREVENTION ACTIVITIES [EMB]

1. Annual Quarantine on Sport-Harvested Mussels

A quarantine is established by the CDPH Director every year from May 1 through October 31 for the purpose of preventing PSP. The annual quarantine also serves to prevent domoic acid poisoning in consumers. The quarantine order (see Appendix D

for the order in English and Spanish, respectively) applies to all species of mussels taken by recreational sport-harvesters. Commercial shellfish harvesters are not affected by this quarantine, but are regulated by conditions and stringent testing requirements set forth in their shellfish growing area certificates and growing area management plans. All county health officers and their representatives are instructed to enforce the quarantine order. Health officers of coastal and bay counties are requested to post warning placards in conspicuous places to advise the sport shellfishing public of the quarantine.

The PSU Chief, in consultation with the Chief of the Disease Investigation Branch, initiates the quarantine order each year. It is forwarded through appropriate channels so that it reaches the Director of CDPH for signature. The approved quarantine order is immediately distributed to all local health officers, directors of environmental health, and to selected authorities in other concerned agencies, such as the Department of Fish and Wildlife, the Department of Parks and Recreation, and the National Park Service. COMMS distributes a press release to all major media outlets and the quarantine order is posted on the CDPH web page. The PSU Chief maintains an up-to-date mailing list for electronic distribution of the quarantine order and for all additional notices or special quarantines.

2. Public Education [EMB]

Public education activities are closely linked with the annual mussel quarantine. When the annual quarantine notice is emailed to County health officers, it is accompanied by a cover letter (Appendix E) that gives instructions concerning quarantine warning placards and outreach to high-risk groups. A document that provides detailed background information on PSP and domoic acid and a summary of the past year's activity is included in this package (Appendix F). These documents, as well as draft CDPH press releases announcing the annual mussel quarantine (Appendix G-1) or special health advisories (Appendix G-2), are prepared by the PSU Chief. Copies of quarantine posters, translated by local health agencies into Spanish, Chinese, three Southeast Asian and two Philippine languages (Appendix H), are provided on request.

Information about shellfish quarantines and health advisories are posted to the CDPH COMMS website, the CDPH Marine Biotoxin Monitoring Website, and available via recording on an information line (800) 553-4133.

The MOU between the CDFW and CDPH (Appendix C) commits CDFW, during the course of normal activities, to increasing the public's awareness of the importance of the annual mussel quarantine. Specifically, CDFW agrees to include information on the quarantine in the "Health Warnings" section of the annual public summary of the California Sport Fishing Regulations. CDFW also posts the information online at: <https://wildlife.ca.gov/Fishing/Ocean/Health-Advisories>.

Some other agencies and institutions also assist in the Marine Biotoxin Monitoring Program by providing public education services that focus on prevention. State and

National Parks personnel post quarantine warning signs issued to them by local environmental health agencies, and may personally warn and deter quarantine violators.

IV. EARLY WARNING SYSTEM

1. Disease Case Reporting [DCDC]

Both PSP (also called "shellfish intoxication" and "mussel poisoning") and domoic acid are classified by CDPH as forms of food poisoning. As such, California regulations (CCR, Title 17, Secs. 2500 and 2574; see Appendix I for a summary of the State laws and regulations relating to the Marine Biotoxin Monitoring Program) require health professionals or any other person attending a known or suspected PSP or ASP case to report it immediately to the local health authority. The local health authority, in turn, must report these cases within seven days to the CDPH Division of Communicable Disease Control. A reminder of this requirement is included in the cover letter to the annual quarantine notice. DCDC in turn notifies the PSU Chief immediately upon receiving a report of a PSP or ASP case so that follow-up shellfish sampling and other surveillance activities can be instituted.

2. Shellfish Monitoring Program [EMB]

Sampling and testing of shellfish meat samples is the only reliable method, other than case reporting, to verify and document PSP, ASP, and DSP activity.

a. Commercial shellfish sampling

CDPH utilizes a routine sampling strategy for the commercial growing areas. As a condition of their shellfish growing area certificates issued by CDPH, commercial shellfish harvesters in each growing area are required to submit at least one shellfish sample per week from locations designated in the Management Plan to MDL for PSP testing during all harvesting periods. Santa Barbara growing area is required to submit samples for domoic acid testing weekly. At Humboldt Bay, Tomales Bay, and Morro Bay growing areas, designated harvesters are required to submit weekly samples, while other harvesters in the same growing area submit samples only when directed to do so by CDPH. CDPH has established phytoplankton thresholds for *Dinophysis* to trigger when DSP testing may be needed in Section IV.3.c.(1). See Appendix J for a list of sampling requirements for commercial shellfish harvesters.

b. Coastal shellfish monitoring

(1) Location of sampling stations

Since most of the commercial shellfish growing areas in California are located within enclosed bays and estuaries, and phytoplankton blooms associated with PSP toxin production appear to originate offshore, the most effective strategy for monitoring PSP

toxicity is to sample shellfish regularly at the outer coast locations. This protects sport-harvesters and provides an early warning to the commercial harvesters. A large number of coastal locations have been sampled over the years for PSP, from which CDPH has selected a set of "key" sampling stations (Appendix K). These stations were selected on the basis of three criteria: (1) they are the stations where past experience has shown that shellfish samples are most likely to give the earliest warning of PSP toxin accumulation, or (2) they are in good locations to give early warning to certified commercial shellfish harvesting areas or popular sport harvesting areas, and (3) the sampling site is accessible and volunteers can collect the shellfish safely.

(2) Indicator Species

The "indicator species" selected for sampling is the California sea mussel (*Mytilus californianus*), although the bay mussel, also known as the Mediterranean mussel (*M. galloprovincialis*, formerly *M. edulis*) is the mussel most often submitted by commercial harvesters, and are also used at some of the sentinel mussel stations (see section 2.b.5.b.).

(3) Samplers

Historically, the lead agencies involved in coastal shellfish sampling have been the environmental health departments of the fifteen coastal counties (and one city, Long Beach). Generally, the local environmental health specialists have routinely conducted this work. Many local environmental health agencies now are critically short-handed, however, so in practice routine coastal shellfish sampling often is conducted by citizen volunteers and staff of a wide variety of other agencies that have been contacted and trained by state and county health personnel. These additional samplers include staff and students of the University of California Santa Cruz, University of California Santa Barbara, and occasionally federal and state park staff. State CDPH shellfish program staff also collect samples at critical locations and assist local agencies when sufficient local staff are not available.

(4) Sampling Frequency

Samplers are requested to achieve a sampling frequency of at least two samples per month from each key station during the months of May through October, and at least one sample per month during the months of November through April. When either PSP toxins or domoic acid are detected in shellfish, efforts are made to increase the frequency of sampling to allow better characterization of the event. Secondary stations may also be employed during this period of increased surveillance. Each harmful algal bloom is unique: the time of onset, rate of increase, geographic extent of the initial and subsequent impact, and duration are all factors that can vary among events. Frequent monitoring of shellfish, in conjunction with additional sources of information and data discussed below, is therefore essential to help guide the appropriate response action.

As discussed earlier, the majority of PSP-related illnesses historically have occurred during the period between May and October. However, phytoplankton observations and domoic acid monitoring have detected blooms of the domoic acid producing diatoms in spring, summer, and fall. Therefore continuous surveillance is of increased importance outside of the annual quarantine period. Frequent monitoring of phytoplankton species composition and relative abundance along the coast is used for early detection of toxigenic species, which in turn allows the program to focus its shellfish sampling efforts, increasing the frequency of sampling in the potentially affected areas.

(5) Sampling Methods

A protocol for field collection and preparation of samples is provided to all samplers by EMB (Appendices L-1, L-2).

(a) Natural mussel populations

Almost all coastal shellfish sampling consisted of collecting sea mussels and bay mussels from natural beds. Although this approach has been augmented by the establishment of sentinel mussel stations [see (b) below], it remains the mainstay of the coastal shellfish monitoring program.

For safety reasons, sampling of natural mussel beds should be performed by at least two persons. Samplers must be very careful in the selection of sites that are not dangerously exposed to heavy surf, and in the timing of their visits to coincide with sufficiently low tides (preferably "minus tide" periods) and safe wind and wave conditions. They must be alert for "sneaker" waves. Samplers must always put personal safety first.

(b) Sentinel mussel stations

A second technique for sampling coastal locations is the placement of mussels in retrievable nylon mesh bags. These are hung by rope from a dock, float or other suitable structure. The location must provide sufficient security to deter tampering by the public. A "do not disturb" sign usually is posted. Use of this technique is somewhat limited because most coastal locations do not have structures that are suitable for this use. At sites where it can be used, it offers the advantage of permitting the safe collection of samples by one person at almost any time, independent of weather and tidal conditions.

Mussels for the sentinel bag samples have been obtained by the following means: bay mussels have been donated by shellfish harvesters from their shellstock and from natural sets of mussels on their structures and product; and sea mussels have been collected from natural beds, usually at the same time those beds are sampled for PSP analysis. Mussels about two to four inches in length are sought, and several pounds are tied in each sample bag.

3. Phytoplankton Monitoring Program [EMB]

As mentioned previously, early detection is key to mobilizing and focusing additional sampling and analytical resources for plankton, shellfish, and other species in the affected region. As a result of the continued efforts of the participants in the CDPH volunteer phytoplankton monitoring program, it has become possible to detect and track numerous harmful algal blooms (HAB) over the past several years. The advance warning provided by frequent phytoplankton sampling and qualitative observations of species composition and relative abundance allows the Marine Biotoxin Monitoring Program to initiate domoic acid analyses or DSP analysis for shellfish and other potentially affected species in the region of highest risk.

a. Commercial shellfish growing areas

Representative phytoplankton samples from each of the state's commercial shellfish growing areas are voluntarily submitted weekly to the Marine Biotoxin Monitoring Program. These samples are screened for the presence of toxin producers to help alert program staff and harvesters of a potential bloom.

b. Coastal monitoring

(1) Location of sampling stations

Since most of the commercial shellfish growing areas in California are located within enclosed bays and estuaries, and phytoplankton blooms associated with PSP and domoic acid toxin production appears to originate offshore, the most effective strategy for marine biotoxin monitoring is to sample shellfish and phytoplankton regularly at the outer coast locations. This protects sport-harvesters and provides an early warning to the commercial harvesters.

A large number of coastal locations have been sampled over the years for HAB phytoplankton species, from which CDPH has selected a set of "key" sampling stations. These stations are selected based on the basis of three criteria: (1) they are the stations where past experience has shown that shellfish samples are most likely to give early warning of biotoxin accumulation in conjunction with a HAB as observed in phytoplankton sampling, (2) they are in good locations to give early warning to certified commercial shellfish harvesting areas or popular sport harvesting areas, and (3) they are accessible and safe for the phytoplankton sampler volunteers.

(2) Samplers

At present the program consists of over 70 participants statewide, representing a wide variety of agencies, organizations, and citizen volunteers. It should be noted that, as with any volunteer-based sampling program, participation and longevity is variable amongst volunteers and periodic recruiting and training is necessary to provide continuity on a regional basis.

Each participant has been trained and equipped to collect phytoplankton samples at one or more locations. A small number of participants have received additional training and equipment for performing field identifications of phytoplankton, with an emphasis on the toxin producing species. The majority of program participants ship their samples to the program office in the CDPH Richmond Laboratory Campus where the species identifications are made using standard light microscopy techniques. All field collection data, field observations, and species identification and relative abundance information is entered into a relational database. Monthly reports are distributed to program participants and other interested persons.

In addition to the efforts of program participants, independent and more quantitative phytoplankton monitoring and research is conducted by a number of colleges, universities, and research organizations in California. Several of these groups informally share information with the Marine Biotoxin Monitoring Program (e.g., via email updates) and alert the program coordinator when a potential toxic bloom is detected. In turn the program provides updates on phytoplankton observations and toxin analysis results throughout the region to the involved individuals in these organizations.

(3) Sampling frequency

Sampling frequency varies among program participants, typically ranging from one sample per week to one sample per month at each site. In an area where sampling effort is lower, efforts are made to train several volunteers to collectively provide a greater frequency of sampling. In general the frequency of phytoplankton sampling along the coast is greater than the frequency of shellfish sampling because of fewer limitations. Access to phytoplankton sampling locations (e.g., piers) is typically easier and safer. In addition, phytoplankton sampling can usually be conducted regardless of tides and weather. The greater frequency of phytoplankton sample collection provides better resolution of data used for establishing and following trends in the distribution and abundance of toxin producers.

c. *Phytoplankton sample analysis*

A cost-effective qualitative approach to phytoplankton monitoring in key areas can be an effective tool for early detection of potential toxic blooms, providing a valuable supplement to the existing shellfish monitoring program. The volunteer phytoplankton monitoring program initiated by CDPH in 1993, with support from the U.S. Food and Drug Administration's Office of Seafood, has proven to be of immense value in providing near real time information on phytoplankton assemblages along the coast. Numerous occurrences of *Alexandrium* and *Pseudo-nitzschia* have been detected prior to the onset of toxicity.

(1) Indicator species

As discussed in Section I, there are specific species of phytoplankton that can produce the toxins of concern. *Alexandrium* sp. is the indicator species for PSP toxins, *Pseudo-nitzschia seriata* complex is the indicator species for ASP and domoic acid, and *Dinophysis* sp. is the indicator species for DSP. The phytoplankton program looks for these species/genera in the phytoplankton samples described in (3) below.

Data from phytoplankton sampling and observations are used in a number of ways. General trends in the distribution and abundance of major groups of species (e.g., diatoms and dinoflagellates), in conjunction with general patterns in oceanographic conditions (e.g., upwelling, downwelling), provide insight into the relative risk of occurrence for each known toxic species. Diatoms such as *Pseudo-nitzschia* are most common during periods of upwelling and dinoflagellates like *Alexandrium* are more likely to occur in elevated densities during relaxation and downwelling periods. Data on the distribution and relative abundance of these specific toxin producers provides insight into the potential magnitude of an impending toxic bloom. This information can provide guidance for determining the region of concern for increased sampling of shellfish and other at-risk species. The ability to knowledgeably focus sampling efforts is a valuable management tool that benefits the state monitoring program staff, the staff of the support laboratories, and the program participants that contribute their time and effort to collect and ship additional samples.

i. *Dinophysis* threshold for DSP testing in commercial growing areas

The EMB has established the following threshold of 34,500 cells per liter for *Dinophysis* concentrations in representative phytoplankton samples that when exceeded will trigger the request for testing of shellfish from the growing area for DSP toxins. Staff of the EMB will prescreen phytoplankton samples with the cell biomass estimate of a relative abundance index (RAI) to determine when total *Dinophysis* cells should be quantified in cells per liter. When the calculated RAI for total *Dinophysis* spp. exceeds 0.056 then EMB will require a whole water sample be collected. EMB will conduct the cell counts on whole water samples for total *Dinophysis* to determine if the 34,500 cells per liter threshold is exceeded.

(2) Sampling methods

A protocol for field collection and preparation of samples is provided to all samplers by EMB (Appendix L-3). A sample submission form accompanies each sample, providing information on the date and location of collection, as well as information on the sampling effort (depth of tow, number of repetitive tows at this depth, and the resultant total tow length). By tracking sampling effort relative to species relative abundance it is possible to follow both temporal and spatial trends in phytoplankton species along the coast.

(3) Light microscopy methods

Phytoplankton samples collected and shipped by volunteers and harvesters to EMB are examined by EMB staff for HAB presence and general phytoplankton species

composition (Appendix L-4). Summarized, this method calls for estimating the percent composition of each phytoplankton species and placing it within a percent range category as follows: Rare (< 1%), Present (1 – 9 %), Common (10– 49%), or Abundant ($\geq 50\%$).

HAB species are scrutinized because the estimation of their increased abundance frequently leads to increase biotoxin monitoring effort with shellfish samples. For example, *Alexandrium* spp. “present” or higher in composition can foretell a PSP event, particularly if the sample is dense with a high settled volume of phytoplankton. Similarly, *Pseudo-nitzschia* in “common” levels of 20% or higher can often foretell a domoic acid event. In both of these cases, if EMB staff were to see these HAB species in these densities or greater, shellfish samples would be requested from volunteers or harvesters that are in the vicinity of where the phytoplankton sample(s) indicated there is a potential bloom, and they would be tested for the relevant biotoxin to determine if any closures or advisories are warranted.

4. Environmental Observations

Qualitative observational data can be quite valuable in providing an early warning to a potential toxigenic phytoplankton bloom. Because marine mammals such as the California sea lion often forage offshore they may be exposed to dangerous levels of biotoxins like domoic acid before nearshore monitoring efforts can detect them. The Marine Mammal Center responds to strandings of marine mammals along large portions of California and records if the symptoms are consistent with domoic acid poisoning. Offshore blooms of *Pseudo-nitzschia* have caused elevated domoic acid concentrations in small planktivorous finfish such as anchovy and sardine.

Generally speaking diatoms are most commonly associated with upwelling conditions, with dinoflagellates prevailing when coastal waters are more stable and warmer. North or northwest winds drive the upwelling process. When these winds decrease for an adequate period of time, upwelling declines and a relaxation event may follow, perhaps followed by downwelling associated with southerly winds. Relaxation and downwelling is associated with the advection of warm water from offshore to the nearshore region. The movement of this warm water onshore may serve as the mechanism by which dinoflagellates are concentrated and delivered to the nearshore area. If *Alexandrium* is part of this dinoflagellate assemblage, an increase in PSP toxicity in shellfish would be expected. A relationship between these physical processes and the onset of PSP toxicity has been previously demonstrated along the west coast of Spain (Fraga, 1988). Almost all (over 99 percent) of the PSP illness cases in California have occurred during the months of May through October. This time period is associated with strong upwelling and relaxation events along the central and north coast of California. However, there is a tremendous amount of information and understanding that is still needed before true predictive tools for biotoxin events can be developed.

V. LABORATORY RESOURCES

1. PSP laboratory testing

Samples submitted to CDPH MDL for PSP toxin analysis. The procedure for mussel samples is: samples are first screened for PSP toxins on Neogen Reveal 2.0 lateral flow assay and only positive samples are tested for PSP toxins by the standard AOAC mouse bioassay NISSP Approved method. If the sample matrix is not a mussel or no rapid test kits are available, the laboratory will test the samples with the AOAC mouse bioassay. Commercial bivalve shellfish reopening samples are tested for PSP toxins by the standard AOAC mouse bioassay.

MDL is the only laboratory in California in conformance with the NISSP standards for PSP toxin testing to perform the AOAC mouse bioassay method. The approximate limit of detection for the mouse bioassay as performed in MDL is 32 µg/100 g of shellfish meat.

2. Domoic acid laboratory analysis

Samples submitted to CDPH FDLB for domoic acid analysis are screened with an in-house high-performance liquid chromatography method (Dhoot, 1993; FDLB, FM-310). When a commercial growing area is closed due to domoic acid, the reopening samples are analyzed using the NISSP approved Quilliam method (Quilliam, 1991). The reporting limit is currently 2.5 µg/g (µg/g = parts per million ([ppm])). The domoic acid federal alert level is 20 µg/g for all shellfish species except for crab viscera (a crustacean), which has a limit of 30 µg/g.

Domoic acid analyses for coastal samples and low-risk commercial shellfish growing areas are conducted as-needed based on phytoplankton monitoring observations of trends in *Pseudo-nitzschia spp.* distribution and relative abundance along the coast. The high-risk growing area in Santa Barbara is monitored at least weekly, by submittal of a sample for HPLC screening for domoic acid. Ancillary information from environmental observations (e.g., reports of marine mammal strandings with animals exhibiting symptoms consistent with domoic acid poisoning) may also trigger the analysis of shellfish samples for domoic acid.

3. DSP laboratory analysis

CDPH does not have the capability to test samples for DSP toxins. Only when the phytoplankton samples indicate elevated levels of *Dinophysis* as described in Section IV. 3.c.(1) or there is an outbreak of DSP will samples be tested for DSP. Samples for DSP testing would go to a laboratory in conformance with the approved NISSP method for DSP. The NISSP approved method for DSP is an LC-MS/MS method. When the need for laboratory support for DSP testing arises, CDPH will identify available support. At the time of publication of this report, possible options identified were: FDA, state of Washington Department of Health Laboratory, and Bieglow Laboratory for Ocean Sciences in Maine.

4. Phytoplankton analysis

Phytoplankton water samples are submitted to the CDPH PSU laboratory for light microscopy analysis as described in Section IV.3.c. There is no NSSP approved method for phytoplankton analysis.

VI. ALERT RESPONSE

1. Alert Notification to Harvesters and Agencies [EMB]

Upon receipt of a shellfish poisoning case report or a laboratory report of an elevated level of PSP toxins or domoic acid in shellfish, EMB will conduct a telephone and/or email notification to alert commercial shellfish harvesters and public agencies (including local environmental health agencies and poison control centers) in the area(s) involved. FDB and CDFW also will be notified. EMB will maintain an emergency notification contact list for this purpose. If an alert occurs during the annual mussel quarantine, local environmental health agencies will be instructed to ensure that quarantine posters are still in place, and CDFW and State and National Parks will be instructed to warn any persons found disregarding the quarantine. A decision tree illustrating the communications and response system for marine biotoxin events is provided in Appendix M.

2. Follow-up Sampling [EMB]

Upon receipt of a shellfish poisoning case report or a laboratory report of an elevated level of PSP toxins or domoic acid in shellfish, EMB will contact samplers and request samples to determine the severity and dimensions of the outbreak. EMB will maintain a contact list of routine and supplemental samplers for this purpose.

EMB may request follow-up samples from adjacent locations in addition to the routine key monitoring stations, and may request an increase in sampling frequency and sampling of additional species. Follow-up sampling in certified commercial shellfish growing areas will include all shellfish species that are commercially harvested in that area. Samplers will identify follow-up samples as such on their laboratory submission slips. MDL will telephone or email the results of all follow-up PSP or domoic acid analyses to the PSU staff as soon as possible.

3. Control Actions

a. Control of Certified Commercial Harvesters

i. Growing Area Protocols for Toxins Below Closure Level [EMB]

Because of the potential for a rapid rate of increase documented for both domoic acid and the PSP toxins in shellfish growing areas, EMB has developed a tiered alert stage system. The tables in Appendices N-1 and N-2 outline multiple alert stages for

concentrations of biotoxins below the closure alert level for PSP and domoic acid, respectively. The tables identify harvest criteria and sampling actions for each stage of toxin concentration. Increasing harvest control actions are implemented as toxin concentration approaches the closure level.

Routine weekly monitoring is the “No Alert” tier. All stages over No Alert call for increased sampling. When low levels of toxins are detected under “1st Stage” (detectable – < 50 µg/100 g PSP toxins or 2.5 - 4.9 ppm domoic acid) harvest is restricted to a sample window of within 48 hours from the sample date and time. Under “2nd Stage” (≥ 50 - 60 µg/100 g PSP toxins or 5 - 9.9 ppm domoic acid) harvest is restricted to a sample window of within 24 hours from the sample date and time. Under “3rd Stage” (≥ 60 - 79 µg/100 g PSP toxins or 10 - 19.9 ppm domoic acid) harvest is restricted to lot release. Lot release means that the harvester must coordinate with CDPH to submit samples from each harvested lot of shellfish and can only release the lot when CDPH has verified its safety based on satisfactory sample results. As described in the Tables in N-1 and N-2, “4th Stage” (≥ 80 µg/100 g PSP toxins or ≥ 20 ppm domoic acid) results in a closure of the growing area. Additional information on closures is described in Section ii. below.

When there is an increase in alert stage, the increase will be effective on the date and time that samples are collected. Alert stages below the closure level will be decreased effective on the sampling date and time after the sample results are received from the laboratory. To be considered for a decrease in alert stage, successive samples collected at least three (3) days apart (i.e., 72 hours) must meet the toxin level requirements for that stage.

ii. Growing Area Closure Orders [EMB]

Closure of a commercial shellfish growing area due to an elevated concentration of marine biotoxin is accomplished by a closure order issued by the PSU Chief or designee. A closure order may be given by telephone and/or email, followed by a letter to the commercial harvester. It is effective until rescinded.

(a) Closure Criteria

(1) Closure alert level.

In compliance with the NSSP Model Ordinance (2023), the closure level for PSP toxins is reached when the concentration of the toxins in shellfish meat equals or exceeds 80 µg/100 g meat. The closure level for domoic acid in bivalve shellfish is 20 ppm. A biotoxin concentration at or above the respective closure level in one or more samples of commercial product in the growing area prior to harvest, or in commercial channels after harvest, will trigger an immediate closure of the growing area until follow-up sampling confirms that the toxin has returned to safe levels (see section 3.a.2.c-d.).

(2) Size of closed area.

The initial closure of a commercial shellfish growing area will include the entirety of the certified areas within the bay, estuary, or body of water in which the growing area is located, unless otherwise described in the current growing area Management Plan. The size of the closure zone may be reduced if current sample data is available from adjacent areas (greater than two miles away from the affected area) that demonstrate that region is below the closure level for the toxin(s). The closure order may be modified later to a smaller area pending the results of follow-up sampling and evaluation of data from additional sites. In reopening any commercial area, or any part of that area located within two miles by water of any site where toxin in a shellfish sample was detected above the closure level, the criteria in paragraph (3.a.2.d.) apply.

(3) Species restrictions.

A closure of a commercial growing area may be limited, at the discretion PSU staff, to the shellfish species determined to have reached the closure level, or it may include all shellfish in the growing area.

(b) Enforcement and Patrol

Responsibility for patrol activity to enforce a growing area closure is shared by FDB and CDFW. Pursuant to the MOU between CDFW and CDPH (see Appendix C), PSU will notify both entities of any growing area closures and coordinate the patrol of commercial shellfishing areas to ensure compliance. Harvesting during a growing area closure by a certified commercial harvester is a serious violation and may result in suspension or revocation of their growing area certificate.

(c) Sampling during a Marine Biotoxin Closure

Follow-up sampling will be continued throughout a closure. All commercially harvestable shellfish species in the growing area will be sampled unless otherwise specified by the PSU Chief. The PSU Chief or their designee will determine sampling sites and times. Sites will be selected so that no part of any certified growing area is more than two miles by water from a sampling site for the species of shellfish harvested in that growing area. To be considered for reopening, successive samples will be collected at least three (3) days apart (i.e., 72 hours).

(d) Reopening Criteria

The closure of a certified shellfish growing area will be terminated separately for each commercially harvestable shellfish species upon the finding that two successive samples of that species, taken over a period of at least three (3) days (i.e., 72 hours) from all sampling stations, as determined by the PSU Chief, within the growing area or within two miles of the growing area by water, are below the closure level of the toxin, and that toxin levels in shellfish from any adjacent but more distant areas are undetectable or exhibit a trend of decreasing concentrations. The successive

samplings will consist of samples randomly selected from the harvestable shellfish species within the certified growing area. All samples must be below the closure level before a closure is terminated.

(3) Identification of Potentially Contaminated Products or Lots - Notification to FDB
[EMB]

The PSU Chief will notify the Chief of the Food Safety Unit, FDB, immediately when any reported case of shellfish poisoning or shellfish sampling results indicate alert levels of PSP toxins or domoic acid in or near a commercial shellfish growing area.

Notification will include the following:

- (a) Type of shellfish
- (b) Sampling date
- (c) Type and level of toxin
- (d) Area potentially affected
- (e) Type of shellfish grown and proximity to positive sampling site(s)
- (f) Last date of any commercial harvest, if known
- (g) Quantity harvested and date distributed, if known
- (h) Closure location, the date and time closure began, and estimate of the length of time closure will be in effect

(4) Harvested Product Actions [FDB]

Actions will be taken in accordance with the criteria in FDA Compliance Policy Guide 7108.20, Chapter 8 -- Fish and Seafood (see Appendix O).

(a) Identification of potentially contaminated products or lots -- notification by FDB

When any harvested product action is warranted, the FDB Food Safety Unit Chief will immediately notify the affected local environmental health agencies and FDA, providing the information in the notification from EMB, and request their assistance for follow-up actions as necessary.

(b) Tracing distribution of potentially contaminated products.

FDB, with assistance from the local environmental health department(s) and FDA, will determine as soon as possible the current location of any potentially contaminated product(s) and request that they be held, pending appropriate regulatory action. If contaminated products have been shipped, FDB will endeavor to determine their distribution within 24 hours of notification. Copies of written records showing the type, quantity, date harvested, harvest location, date sold, and name and address of the consignees, will be obtained from the packer/distributor.

(c) Embargoing adulterated shellfish already on the market.

In accordance with Section 26830 of the California Health and Safety Code, FDB will establish embargoes, as necessary, at the processor, packer, repacker, distributor, restaurant or retail market.

(d) Recalling shellfish already purchased by consumers.

Recalls, classified as Type I, II or III, will be conducted according to the FDB Policy Guide for Complaints, Tampering and Recalls (Appendix P).

(e) Suspending or withdrawing interstate certification.

If necessary, the Chief of the Food Safety Unit, FDB, will suspend or withdraw a dealer's certificate and advise FDA, Seattle, to remove the dealer from the Interstate Certified Shellfish Shippers List (ICSSL). *(NOTE: This procedure must be provided for under the Model Ordinance guidelines, but in actual practice FDB never has found it necessary for control of PSP in California.)*

b. Control of Recreational Sport-Harvesters [EMB]

i. Health Advisory

The PSU Chief, EMB, will initiate a press release containing a health advisory to the public whenever phytoplankton and shellfish monitoring results indicate the development of a potentially serious toxic event (see example, Appendix G-2). The press release will be forwarded through administrative channels as rapidly as possible for review and ultimate approval by the CDPH Director and dissemination by COMMS.

(a) Health advisory criteria.

(1) Alert level and number of samples required.

The alert level for PSP is reached when the concentration of PSP toxins in shellfish meat equals or exceeds 80 µg/100 g of meat. The alert level for domoic acid toxicity is greater than or equal to 20 µg/g of shellfish meat (i.e., 20 ppm). When found in one or more coastal shellfish samples, the alert level of toxin will trigger immediate follow-up sampling and evaluation of the need for special quarantine action. If the toxin concentration is already at a dangerous level such that there is not adequate time for additional sampling, CDPH may immediately issue a health advisory if the public's health is at risk.

(2) Size of affected area

Health advisories are typically applied on a County by County basis. A safety zone may be added at the discretion of the PSU Chief or the EHSS Chief to extend the quarantine limits up to 50 miles or to the next county line. A special case is Monterey Bay, which

can be impacted by high toxin concentrations while open ocean sites in Santa Cruz and Monterey counties are unaffected.

(3) Species covered

A health advisory will include all species of mussels, and any other shellfish species that have reached the alert level; or it may, at the discretion of the PSU Chief or the EHSS Chief, include all bivalve shellfish species found in the quarantine area. Additionally, other seafood species like small finfish (e.g., anchovy and sardine) and crustaceans (crab and lobster, particularly their viscera) may contain dangerous levels of toxin and can be included in the health advisory.

(4) Other instructions

- a. If a health advisory for PSP is limited to mussels, the press release will include an advisory to remove and discard the viscera of scallops and clams as well as the dark (pigmented) parts of the siphons of Washington clams (also known as butter clams, *Saxidomus nuttalli*).
- b. Local health officers will be instructed to post notices of the health advisory at access points within the affected area and will ensure that the area remains posted until the advisory is lifted.

(b) Notification Distribution

All potentially affected persons, companies and agencies on the CDPH emergency notification list will be notified immediately by email or telephone as soon as a health advisory has been issued. The health advisory press release will be sent to all local health officers and directors of environmental health, and to designated contact persons in the CDFW and any other concerned agencies in the quarantined area.

(c) Enforcement and patrol

Local environmental health agencies have lead responsibility for enforcement of all shellfish poison health advisories within their jurisdictions, and for the posting of signs at appropriate locations to warn the public of the hazard. In cooperation with the local environmental health agencies, the following agencies also may perform enforcement and posting within their jurisdictions: city or county parks departments, the State Department of Parks and Recreation, the National Park Service, and the United States Forest Service. In addition, pursuant to the MOU between CDPH and CDFW, CDPH will notify CDFW of any special quarantines and CDFW will increase coastal patrols near popular shellfishing areas and will provide information or warnings to the public as necessary.

(d) Sampling

Follow-up sampling of mussels and phytoplankton will be continued in areas subject to a health advisory on sport-harvested shellfish. Additional sampling should be conducted in areas adjacent to the quarantine area to assist in tracking the bloom. Successive samples will be collected as frequently as practical to maintain information on bloom intensity and distribution. Sampling sites should be spaced approximately five to 15 miles apart, depending on the availability of suitable sampling sites and the pattern of use of the area by sport harvesters. Where a shellfish-poisoning outbreak affects areas where clams and cockles may be harvested, those species also should be sampled. CDPH typically relies on the local expertise of CDFW biologists to assist in providing clam samples.

(e) Termination criteria

With the exceptions noted below, a health advisory on sport-harvested shellfish shall be rescinded upon the finding that at least two successive mussel samples taken over a period of at least 14 days are below the alert level of the toxin at all sampling sites that had elevated results in the affected area. The same criteria shall apply to other seafood species included in the health advisory. At the discretion of the PSU chief or EHSS chief, results from an adjacent sample site can be used to rescind an advisory if samples from the elevated site are unavailable.

Exceptions: At the discretion of the PSU Chief or the EHSS Chief, individual shellfish species may be removed earlier from a special advisory upon the finding that at least two successive samples of that species taken over a period of at least 14 days from sampling sites in all major sport-harvesting areas for that species within the advisory area are below the alert level.

Exception: PSP and Washington clams

Upon the lifting of a PSP health advisory for mussels, the health advisory may be extended for Washington clams (also known as butter clams, *Saxidomus nuttalli*) due to their ability to retain these toxins for very long periods of time. An additional factor in the extension is that this species is seldom sampled, hence the data typically does not exist to include them in the termination of the health advisory. An extension may be effective for at least six months, directing that the siphons of any Washington clams taken within the former advisory area must be discarded before the clams are presumed safe for human consumption. If the sampling criteria for termination of the advisory are met, then CDPH may rescind the advisory sooner than the six month interval.

Exception: domoic acid and razor clams

Upon the lifting of a domoic acid health advisory for mussels, the health advisory may be extended for razor clams due to their ability to retain this toxin for very long periods of time. An additional factor in the extension is that this species is seldom sampled, hence the data typically does not exist to include them in the termination of the health advisory. An extension may be effective for at least six months. If the sampling criteria for termination of the advisory are met, then CDPH may rescind the advisory sooner than the six month interval.

c. Dissemination of information within CDPH

(1) General [EMB and FDB]

The PSU Chief, EMB, is designated to receive and disseminate all information related to shellfish poisoning. The PSU Chief will inform the EHSS Chief, and the FDB Food Safety Unit Chief immediately upon being notified of any problem related to shellfish poisoning that may require emergency control action. The Chief of EMB will ensure that the Chief, EMB, is kept fully informed at all times of any information relating to emergency control actions. The PSU Chief will also notify the FDB Food Safety Unit Chief in the event that market samples of seafood are required for testing.

(2) Public Information

(a) Primary contacts.

The PSU Chief, EMB, and the Food Safety Unit Chief, FDB, will act as primary public information contacts for information concerning their units' areas of responsibility.

(b) Press release.

The PSU Chief will draft a press release announcing any health advisory. It will be submitted through appropriate channels to the Center for Environmental Health (CEH) for approval and to COMMS for editing and release pending approval by CEH. Upon approval, it will be released by the COMMS. The PSU Chief will distribute the press release to affected County Health Officers and Environmental Health Directors, as well as program participants in the shellfish and phytoplankton sampling efforts.

(c) Targeted high-risk groups.

Local health agencies should maintain current mailing lists of agencies and organizations that serve or represent high-risk groups for targeted distribution of shellfish poisoning-related health education materials and press releases. Ethnic groups currently determined to be at high risk for shellfish poisoning are recently arrived Southeast Asians, including Vietnamese and Filipinos. Local health agencies should distribute copies of the health advisory press release to their targeted high-risk group contacts.

(d) Follow-Up press releases.

Additional press releases may be prepared and distributed when a health advisory remains in effect longer than 30 days, and when it is terminated.

(e) Other contacts.

The PSU Chief and staff, as well as local environmental health agency staff, will respond to the many public inquiries that occur in connection with shellfish poisoning outbreaks.

(f) Public Access: Toll-Free Hotline.

A voicemail system on a dedicated toll-free telephone line (800-553-4133) is maintained for public access. The outgoing message informs the public of existing quarantines and health advisories, indicating the regions and species affected. The public may leave messages on this voicemail system requesting further information from PSU staff. The telephone number for this dedicated line is published in all press releases and quarantine orders.

(g) Public Access: CDPH website.

The Health Advisory and Quarantine notices are posted to the CDPH COMMS website and an interactive map of the advisories are on the PSU Program's website. The PSU website also contains additional information about biotoxins in the form of FAQs and leaflets.

VII. PROGRAM EVALUATION

1. Records of Illnesses [DCDC]

Records of shellfish poisoning illnesses are compiled and maintained by the Chief, DCDC. These records include data on the incidence of illness and appropriate case history data. They are available to the PSU Chief, EMB, and have been used in a comprehensive analysis of the State PSP prevention program (Price et al., 1991). These records are important for defining the extent and severity of the problem, making correlations with the occurrence of blooms, as well as for making a retrospective evaluation of the adequacy of the entire program.

2. Shellfish Sample Results [EMB]

Records of shellfish sampling results and phytoplankton observations are maintained by PSU. These shellfish records include locations and dates of collection, name and agency of sampler, shellfish species sampled, concentrations of PSP toxins and/or domoic acid. Plankton data records include equivalent information on sample collection, as well as the identification of species present and their relative abundance. These data are used to compile reports and annual summaries with analyses of trends. These reports are emailed to an extensive list of State and local environmental health agencies, sampling cooperators, commercial shellfish harvesters and other interested parties and posted on the program web page. A summary of the time span and number of shellfish samples recoded in the PSU database for each growing area is summarized in Appendix A Table A-1.

3. Phytoplankton Monitoring [EMB]

The volunteer-based phytoplankton monitoring program generates over 1700 samples per year, providing invaluable data on the distribution and abundance of toxin-producing phytoplankton along the California coast. This information is summarized in the monthly biotoxin reports described above, providing the opportunity to evaluate the effectiveness of this monitoring effort with respect to the shellfish biotoxin data generated during that time period. Over the years it has been demonstrated that routine phytoplankton sampling and observations at key locations can provide an early warning of potential toxic algae blooms, allowing the shellfish sampling program to focus its efforts in the affected areas. In addition, routine phytoplankton monitoring provides valuable information on potential trends in the composition, distribution and abundance of phytoplankton species along the coast. Frequent phytoplankton observations also provide the opportunity to detect the occurrence of other known toxin-producing phytoplankton species along the California coast (e.g., *Dinophysis acuta*).

4. Preparation of Reports on Shellfish Biotoxins and Toxic Bloom Events [EMB]

The PSU Chief, EMB, prepares ongoing status reports and retrospective analyses of toxic phytoplankton blooms, shellfish toxicity and shellfish poisoning events on an "as needed" basis for distribution as cover memos with the monthly sampling results reports. These reports to the local environmental health agencies, commercial shellfish harvesters, and other cooperating agencies and individuals serve to keep them informed of current trends, as well as providing a reminder of the importance of their continued efforts.

Shellfish toxicity data are also used by staff scientists for analysis of the duration and dimensions of toxic blooms, analyzing differences among shellfish species, and for constructing intoxication and detoxification curves. Phytoplankton abundance data and shellfish toxicity data, in conjunction with remote sensing data on physical parameters (e.g., SST, movement of water masses) are used to investigate potential patterns for the purpose of developing predictive tools for toxic blooms. This work may be presented at scientific meetings, published as separate program technical reports, or submitted for publication in the refereed literature in collaboration with other researchers.

IX. REFERENCES

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- Work, T.M.; Barr, B.; Beale, A.M.; Fritz, L.; Quilliam, M.A.; Wright, J.L.C. 1993. Epidemiology of domoic acid poisoning in brown pelicans (*Pelecanus occidentalis*) and brandt's cormorants (*Phalacrocorax penicillatus*) in California, *Journal of Zoo and Wildlife Medicine*, 24(1): 54-62.

Appendix A. Commercial shellfish growing areas in California.

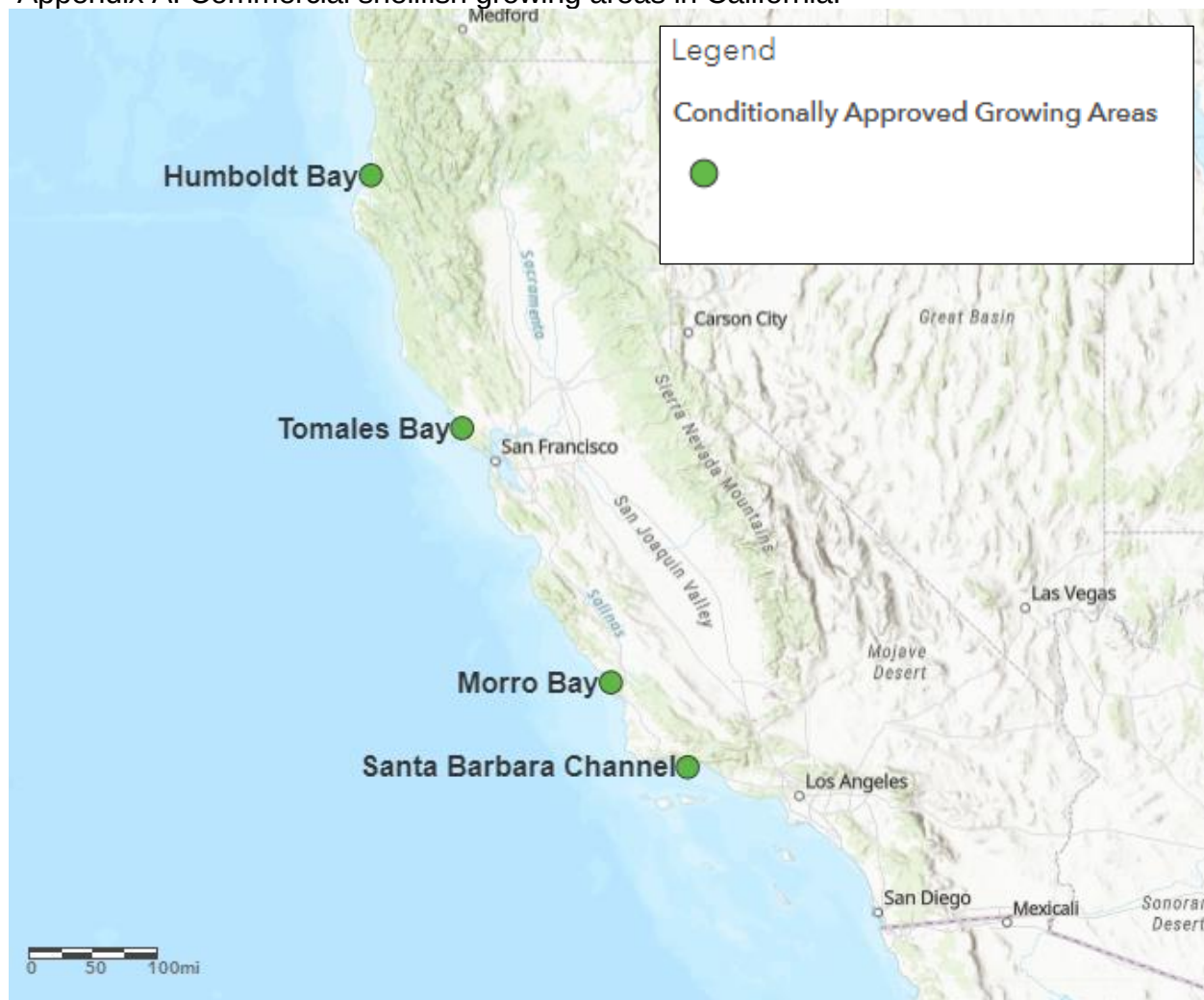


TABLE A-1. Biotoxin database dataset for each commercial growing area.

Growing Area	PSP toxins Dataset		Domoic acid Dataset		Phytoplankton Dataset	
	Year ¹	Number of samples ²	Year ¹	Number of samples ²	Year ¹	Number of samples ²
Humboldt Bay	1982	3,566	1991	667	1993	1,991
Tomales Bay	1981	5,386	1992	680	1993	1,694
Morro Bay	1981	4,252	1992	1,881	1993	1,501
Santa Barbara	2003	1,750	2003	1,259	1993	2,708

¹ Year consistent sampling started.² To March 2023.

Appendix B. CDPH staff involved in the Marine Biotoxin Monitoring Program.

Center For Environmental Health (CEH): June Weintraub, Deputy Director	Division Of Food and Drug Safety (DFDS): Dave Mazzera, Chief
Division Radiation Safety and Environmental Management (DRSEM): Anthony Chu, Chief	Food and Drug Branch (FDB): Benson Yee, Chief
Environmental Management Branch (EMB): Chris Pace, Chief	Food Investigation and Technical Support Section: Michael Needham, Chief
Environmental Health Services Section (EHSS): Eric Trevena, Chief	Food Safety Unit: Christina Grant Shellfish Program Specialist
Preharvest Shellfish Unit Staff (PSU):	
Vanessa Zubkousky-White, Chief Marine Biotoxin Monitoring Program Senior Environmental Scientist (510) 412-4635	Joe Christen Senior Environmental Scientist (510) 229-2822
Sam Rankin Environmental Scientist (510) 412-4633	Matt Scanlon Environmental Scientist (510) 412-4641
Steve Etter Environmental Scientist (510) 412-4631	
EMB Administrative Offices:	
California Department of Public Health 850 Marina Bay Parkway, G165 Richmond, CA 94804	California Department of Public Health P.O. Box 997377 Sacramento, CA 95899-7377
Office of Communication:	
Sacramento (916) 440-7259	

Appendix C. Memorandum of understanding between CDPH and CDFW.

(INSERT CDFW/CDPH MOU: 6 pages)

Appendix D. Annual mussel quarantine order (English and Spanish).

(INSERT APPENDIX D: CURRENT ANNUAL QUARANTINE ORDER – Pages 2)

Appendix E. Annual mussel quarantine cover letter.

(INSERT ANNUAL QUARANTINE COVER LETTER: 3 pages)

Appendix F. Annual mussel quarantine background information.

(INSERT ANNUAL Q BACKGROUND INFO: 6 Pages)

Appendix G-1. Annual mussel quarantine press release.

(INSERT ANNUAL Q NEWS RELEASE: 2 Pages)

Appendix G-2. Special health advisory press release.

(INSERT HEALTH ADVISORY NEWS RELEASE: 2 Pages)

Appendix H. Annual mussel quarantine posters in various languages.

(insert scanned signs for each language: 9 Pages)

Appendix I. Summary of relevant state laws and regulations.

(insert scans of state regs: Pages 9)

Appendix J. Sampling requirements for commercial shellfish harvesters.

All commercial shellfish harvesters are required to submit samples of shellfish from their growing areas for PSP toxin and domoic acid analyses, regardless of the current harvesting status, unless arrangements have been made with the PSU Chief. Certified harvesters are required to conduct weekly phytoplankton sampling. The information from this field-based approach to monitoring has proven valuable for the early detection of a potentially toxic bloom.

In some growing areas there are several companies operating within close proximity to one another. In this special circumstance CDPH has designated certain companies as responsible for submitting weekly samples that are representative of all leases in the vicinity. Additionally, those harvesters not submitting weekly samples under this plan are required to submit samples when instructed to do so by CDPH.

All samples must be shipped to the CDPH Microbial Diseases Laboratory (MDL) at Richmond, **to arrive no later than 10:30 AM on Thursday of each week**, or earlier as directed by PSU. **It is the responsibility of each commercial harvester to ensure that their samples arrive at MDL before the weekly deadline.** MDL assays all samples for PSP toxicity. A portion of each sample may be transferred to FDL for domoic acid analysis as needed. Growing areas considered high risk for domoic acid need to coordinate sample submission with CDPH to assure the sample will also be tested for domoic acid that same week.

Commercial shellfish harvesters shall submit additional shellfish samples when requested by CDPH. Commercial shellfish harvesters may be required to maintain sentinel mussel stations as deemed necessary by the PSU Chief. **Failure to fulfill these sampling obligations may result in (1) harvest closures and/or (2) suspension or revocation of the growing area certificate.** These routine sampling requirements are stated as conditions of their growing area certificates. The following is a summary of the individual sampling requirements.

MARINE BIOTOXIN MANAGEMENT PLAN FOR CALIFORNIA SHELLFISH

Table J-1. Commercial Shellfish Biotoxin Sampling

COMPANY	FREQUENCY ¹	SAMPLE TYPE	LOCATION	DESCRIPTION
Humboldt Bay				
Pacific Shellfish – Humboldt LLC	A	Mussels	Indian Island	Sentinel
	A	Mussels	Channel Marker 13	Sentinel
	B	Oysters, Clams	Harvest Beds	Current Harvest Area
North Bay Shellfish Company	B	Oysters	Harvest Beds	Current Harvest Area
Humboldt Bay Oyster Company	B	Oysters	Harvest Beds	Current Harvest Area
Aqua Rodeo Farms	B	Oysters, Mussels	Harvest Beds	Current Harvest Area
Hog Island Oyster Company	B	Oysters	Harvest Beds	Current Harvest Area
Tomales Bay:				
<i>Outer Bay²:</i>				
Hog Island Oyster Company	A	Mussels	Lease M430-15	Sentinel, Harvest Area
	A-2024; B	Oysters, Clams	Leases M430-15, M430-11, M-430-10	Harvest Area
Marin Oyster Company	A-2025; B	Oysters	Lease M430-02	Harvest Area
Point Reyes Oyster Company	A-2026; B	Oysters, Clams	Lease M430-17	Harvest Area
Charles Friend Oyster Co.	A-2023; B	Oysters, Clams, Mussels	Lease M430-04	Harvest Area

Table continued on next page.

¹ A = weekly sampling; B = supplemental sampling as needed.

² Annual rotational list for submitting weekly oyster sample; other species (e.g., clams) sampled as directed by CDPH.

MARINE BIOTOXIN MANAGEMENT PLAN FOR CALIFORNIA SHELLFISH

Table J-1 continued.

COMPANY	FREQUENCY	SAMPLE TYPE	LOCATION	DESCRIPTION
Mid Bay:				
Point Reyes Oyster Company	B	Oysters, Mussels, Clams	Lease M430-14	Harvest Area
Starbird Mariculture	A	Mussels	Lease M430-06A	Harvest Area
	B	Oysters	Lease M430-06A	Harvest Area
Cove Mussel Company (inactive)	B	Oysters	Lease M430-06	Harvest Area
Hog Island Oyster Company	B	Oysters	Lease M430-12	Harvest Area
Point Reyes Oyster Company	B	Oysters	Lease M430-13	Harvest Area
Marin Oyster Company	B	Oysters, Mussels, Clams	Lease M430-19	Harvest Area
Inner Bay:				
Tomales Bay Oyster Company	B	Mussels, Oysters, Clams	Lease M430-05	Harvest Area
Morro Bay:				
Morro Bay Oyster Company	A	Oysters	Lease M614-01 Parcel 2	Harvest Area
Grassy Bar Oyster Company	B	Oysters	Lease M614-01 Parcel 1 or Lease M614-02	Harvest Area
Santa Barbara:				
Santa Barbara Mariculture Co.	A	Mussels	Lease M653-01	Harvest Area
	A	Oysters	Lease M653-01	Harvest Area
The Cultured Abalone Farm	B	Mussels, Oysters	Intake PRC 7456.1	Wet Storage Intake

¹ A = weekly sampling; B = supplemental sampling as needed.

Appendix K-1 to K-4. Shellfish and phytoplankton sampling stations.

(INSERT Appendix K-1 to K-4: 4 PAGES)

Appendix L-1. Protocol for shellfish sample collection, preparation, and shipping: commercial harvesters.

(INSERT Appendix L-1: 3 PAGES)

Appendix L-2. Protocol for shellfish sample collection, preparation, and shipping: noncommercial.

(INSERT Appendix L-2: 2 PAGES)

Appendix L-3. Protocol for phytoplankton sample collection, preparation, and shipping.

(INSERT Appendix L-3: 2 PAGES)

Appendix L-4. Phytoplankton Sample Observations Protocol.

(INSERT Appendix L-4: 2 PAGES)

Appendix M. Communications and response decision trees.

INSERT APPENDICES M-1 AND M-2: 2 FLOW CHARTS: 2 pages

MARINE BIOTOXIN MANAGEMENT PLAN FOR CALIFORNIA SHELLFISH

Appendix N-1. Management Protocol for PSP in Commercial Shellfish Growing Areas.

ALERT STAGE	PSP TOXIN CONCENTRATION	ACTION¹
No Alert	Nondetectable: < ~34 ug/100g	- Routine Weekly Sampling; - Harvest date must be < 168 hours from sampling date.
1st Stage	Detectable: < 50 ug/100g	- Harvest date must be < 48 hours from sampling date; - Notify CDPH date and time of sample collection and shipping information ² ; - Immediate resample ³ ; - Additional sampling as directed.
2nd Stage	Detectable: < 60 ug/100g	- Harvest date must be < 24 hours from sampling date; - Notify CDPH date and time of sample collection and shipping information ² ; - Immediate resample ³ ; Twice-weekly sampling at minimum ³ ; - Immediate harvest closure if harvest lot laboratory results have not been obtained; - Expand sampling to adjacent growing areas.
3rd Stage	60 - 79 ug/100 g	- Lot test and release ^{3,4} ; - Immediate harvest closure if harvest lot laboratory results have not been obtained. - Immediate resample; Twice-weekly sampling at minimum ³ ; - Expand sampling to adjacent growing areas.
4th Stage	≥ 80 ug/100 g	- Closure of growing area; - Lot test and release ^{3,4} of shellstock from adjacent harvest areas that remain open; - Twice-weekly sampling ³ ; - Expand sampling to adjacent growing areas.

¹ For each alert stage, the alert shall be continued until either (a) a higher stage alert is declared, or (b) two successive samples taken over a minimum period of three days have PSP toxin levels consistent with a lower alert stage and downward trends in biotoxin levels in the surrounding area.

² For growing areas with more than one harvester. Notify CDPH on day of sample collection.

³ Contingent upon arrangement with certified laboratory.

⁴ Lot test and release means that the harvester must submit samples from each harvested lot of shellfish and shall not release that lot to market until CDPH has verified its safety based on satisfactory analytical results. Product must be held under appropriate temperature control while awaiting analytical results. Harvesters should consult the CDPH Food and Drug Branch for guidance on proper handling and storage of the harvested shellstock.

Appendix N-2. Management Protocol for Domoic Acid in Commercial Shellfish Growing Areas.

ALERT STAGE	DOMOIC ACID CONCENTRATION	ACTION ¹
No Alert	< 2.5 ppm	- Routine Weekly Sampling; - Harvest date must be < 168 hours from sampling date.
1 st Stage	2.5 – 4.9 ppm	- Harvest date must be < 48 hours from sampling date; - Notify CDPH date and time of sample collection and shipping information²; - Immediate resample³; - Additional sampling as directed.
2 nd Stage	5 – 9.9 ppm	- Harvest date must be < 24 hours from sampling date; - Notify CDPH date and time of sample collection and shipping information²; - Immediate resample³; Twice-weekly sampling at minimum³; - Immediate harvest closure if harvest lot laboratory results have not been obtained; - Expand sampling to adjacent growing areas.
3 rd Stage	10 – 19.9 ppm	- Lot test and release^{3,4}; - Immediate harvest closure if harvest lot laboratory results have not been obtained; - Immediate resample³; Twice-weekly sampling at minimum³; - Expand sampling to adjacent growing areas.
4 th Stage	≥ 20 ppm	- Closure of growing area; - Lot test and release^{3,4} of shellstock from adjacent harvest areas that remain open; - Twice-weekly sampling³; - Expand sampling to adjacent growing areas.

¹ For each alert stage, the alert shall be continued until either (a) a higher stage alert is declared, or (b) two successive samples taken over a minimum period of three days have domoic acid levels consistent with a lower alert stage and downward trends in biotoxin levels in the surrounding area.

² For growing areas with more than one harvester. Notify CDPH on day of sample collection.

³ Contingent upon arrangement with certified laboratory.

⁴ Lot test and release means that the harvester must submit samples from each harvested lot of shellfish and shall not release that lot to market until CDPH has verified its safety based on satisfactory analytical results. Product must be held under appropriate temperature control while awaiting analytical results. Harvesters should consult the CDPH Food and Drug Branch for guidance on proper handling and storage of the shellstock.

Appendix O. FDA compliance policy guide, 7108.20, Fish and Seafood.

From:
U.S. Food and Drug Administration
Compliance Policy Guide
Chapter 5, Subchapter 540.250

Sec. 540.250 Clams, Mussels, Oysters, Fresh, Frozen or Canned - Paralytic Shellfish
Poison (CPG 7108.20)

BACKGROUND:

Paralytic shellfish poison is a marine biotoxin, produced in single celled marine protozoa of the order Dinoflagellata. This poison accumulates in the tissue of bivalve shellfish that have ingested certain species of dinoflagellates of the genus *Gonyaulax* or closely related genera. Toxin levels in shellfish are associated with algal "blooms" of toxic dinoflagellates in shellfish growing waters of the Pacific Coast and the Bay of Fundy region. Ordinary cooking does not ensure safety of shellfish since the poison is very stable and is not destroyed completely by the usual heat treatments.

REGULATORY ACTION GUIDANCE:

The following represents criteria for recommending legal action to CFSAN/Office of *Compliance*/Division of Enforcement (HFS-605):

Actionable if one sub from a lot shows paralytic shellfish poison value of 80 micrograms, or more, per 100 grams meat when bioanalyzed by current Association of Official Analytical Chemists procedure.

Articles meeting the above criterion represent a potential health hazard. Consequently, recall is the action of choice. Notify CFSAN/Office of *Compliance*/Division of Enforcement (HFS-605) immediately if articles meeting the above criterion are encountered.

Notify CFSAN's Office of Seafood (HFS-415) immediately when articles containing any detectable levels of paralytic shellfish poison are encountered.

Material between asterisks is new or revised.

Issued: 10/1/80

Revised: 10/30/89, 3/95, 5/2005

Updated: 11/29/2005

Appendix P. FDB policy guide for complaints, tampering and recalls.

(INSERT APPENDIX P1-3)